Instance: the Orbit Space is a Charted Space

My first Mathlib PR

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My first Mathlib PR!

PR #35653

feat: add ChartedSpace structure on orbit space #35653

Open pepamontero wants to merge 19 commits into `leanprover-community:master` from `pepamontero:orbit-space-charted`

Conversation 2 | Commits 19 | Checks 10 | Files changed 3

pepamontero commented 3 days ago • edited

feat: add ChartedSpace structure on orbit space

Add topological preliminaries for properly discontinuous group actions, and use them to equip the orbit space `orbitRel.Quotient G M` with a `ChartedSpace` structure.

In particular:

- Show that the quotient map `Quotient.mk _ : M → orbitRel.Quotient G M` is a covering map and hence a local homeomorphism.
- Construct chosen local inverses of the quotient map and establish properties over them.
- Define a `ChartedSpace` instance on the quotient by composing these local inverses with charts of M.

This started as an ItaLean2025 project; here is the Zulip thread dedicated to it.

Big thanks to Michael Rothgang!

Introduction

Goal

Formalize in Lean that the quotient of a manifold by a properly discontinuous group action inherits a natural charted space structure.

Main result

If M is a charted space and G acts properly discontinuously on M, then the quotient space G/M is a charted space.

Contribution to Mathlib

- Show the quotient map is a local homeomorphism
- Construct local inverses of the quotient map
- Define a `ChartedSpace` instance on the quotient space

Prerrequisites

What will we need?

- Topological Spaces
- Partial Functions
- Groups and group actions

Topological Spaces

A topological space is a set X with a definition of what it means for some $U \subseteq X$ to be open, satisfying the following conditions:

```
class TopologicalSpace (X : Type u) where
  /-- Def of what 'IsOpen' means. -/
  IsOpen : Set X → Prop

  /-- X (the universe) is open.-/
  isOpen_univ : IsOpen univ

  /-- The intersection of two opens is open. -/
  isOpen_inter : ∀ s t, IsOpen s → IsOpen t → IsOpen (s ∩ t)

  /-- The union of a family of opens is open. -/
  isOpen_sUnion : ∀ s, (∀ t ∈ s, IsOpen t) → IsOpen (⋃₀ s)
```

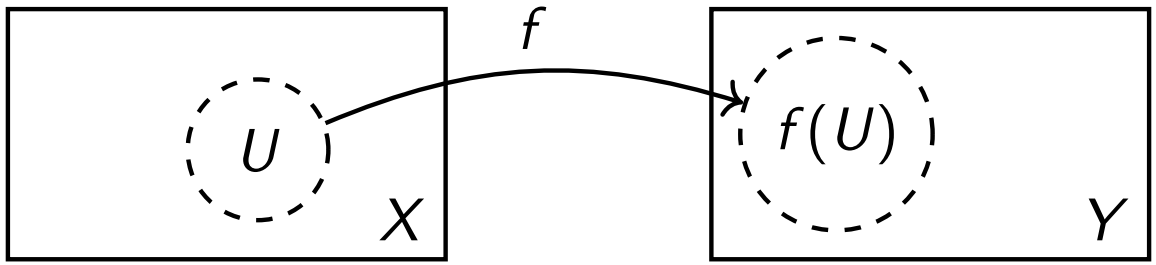
Partial Functions

Main idea:

- We have a function that is bijective locally, e.g.

$$f : U \subseteq X \to f(U) \subseteq Y.$$

- We don't care about f outside of U, it could be undefined even.
- We want to package inside the same object information like:
  - Where f is defined and is bijective
  - What the inverse of f is
  - Where the inverse of f is defined



PartialEquiv

We may define an object `f : PartialEquiv X Y` defined in U with the following objects:

```
structure PartialEquiv (α : Type*) (β : Type*) where
  toFun : α → β -- f
  invFun : β → α -- inverse of f
  source : Set α -- U
  target : Set β -- f(U)

  -- the source and target satisfy:
  map_source' : ∀ x ∈ source → toFun x ∈ target
  map_target' : ∀ x ∈ target → invFun x ∈ source

  -- and invFun is in fact an inverse (locally):
  left_inv' : ∀ x ∈ source → invFun (toFun x) = x
  right_inv' : ∀ x ∈ target → toFun (invFun x) = x
```

OpenPartialHomeomorph

In particular, we have `OpenPartialHomeomorph X Y`, which is a `PartialEquiv X Y` in which

- X and Y are topological spaces,
- U and $f(U)$ are open sets, and
- f and its inverse are continuous functions.

```
structure OpenPartialHomeomorph (X : Type*) (Y : Type*)
    [TopologicalSpace X] [TopologicalSpace Y]
    extends PartialEquiv X Y where
  open_source : IsOpen source
  open_target : IsOpen target
  continuousOn_toFun : ContinuousOn toFun source
  continuousOn_invFun : ContinuousOn invFun target
```

Group action

Definition

Let X be a set and G a group. A *right action* of G on X is a map

$$\begin{array}{rcl} X \times G & \to & X \\ (x, g) & \mapsto & x \cdot g \end{array}$$

such that

1. $x \cdot 1 = x$ for all $x \in X$
2. $(x \cdot g) \cdot g' = x \cdot (gg')$ for all $x \in X$ and $g, g' \in G$

- It is *continuous* if the map $x \mapsto x \cdot g$ is continuous.
- It is *free* if for each $x \in X$, if $g \in G$ and $g \cdot x = x$ then $g = 1$.
- It is *properly discontinuous* if for every compact set $K \subset X$ there are only finitely many $g \in G$ with $(g \cdot K) \cap K \neq \emptyset$.

Group action: example

Example

The action

$$\begin{array}{rcl} \mathbb{R} \times \mathbb{Z} & \to & \mathbb{R} \\ (x, n) & \mapsto & x + n \end{array}$$

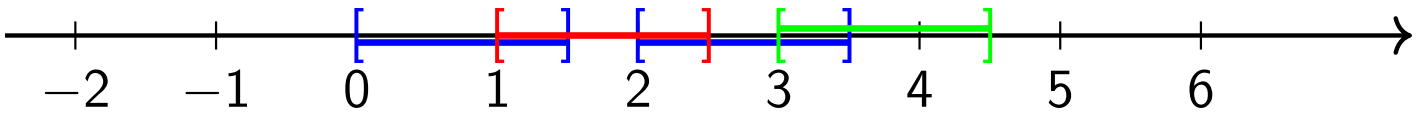
is a (right) group action. Indeed:

1. $x + 0 = x$
2. $(x + n) + m = x + (n + m)$

It is a properly discontinuous action: if $K \subseteq [a, b]$ is a compact, then the intersection:

$$[a, b] \cap [a + n, b + n]$$

can only be nonempty for finitely many $n \in \mathbb{Z}$.



Orbit Space

Definition

The *quotient* (or *orbit space*) X/G is the set of G-orbits of X, with the *projection map*

$$\begin{array}{rcccl} \pi & : & X & \longrightarrow & X/G \\ & & x & \longmapsto & [x]_G = \{x \cdot g | g \in G\} \end{array}$$

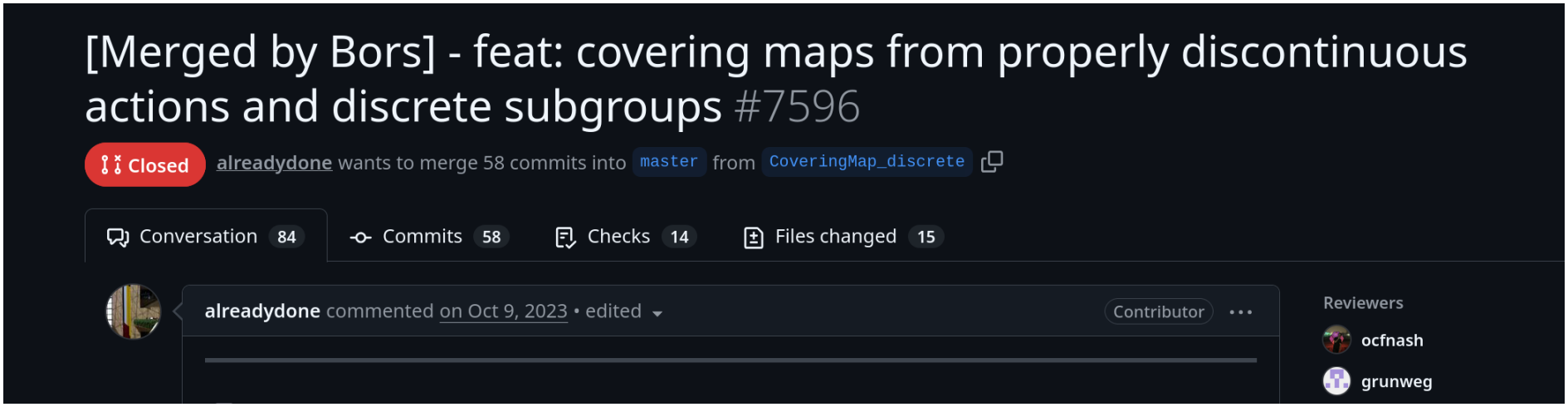
```
MulAction.orbitRel.Quotient (G : Type u_1) (α : Type u_2)
  [Group G] [MulAction G α] : Type u_2
```

Example

$$\begin{array}{rcccl} \pi & : & \mathbb{R} & \longrightarrow & \mathbb{R}/\mathbb{Z} = \mathbb{S}^1 \\ & & x & \longmapsto & [x]_{\mathbb{Z}} = \{x + n | n \in \mathbb{Z}\} \end{array}$$

PR #7596

Our project depended on PR #7596, which was closed on Dec 23, 2025.



It added two things we're interested in:

```
lemma isQuotientCoveringMap_quotientMk_of_properlyDiscontinuousSMul
  [...] [ProperlyDiscontinuousSMul G E]
  : IsQuotientCoveringMap (Quotient.mk (orbitRel G E)) G

lemma isQuotientCoveringMap.isCoveringMap
  (h : IsQuotientCoveringMap f G) : IsCoveringMap f
```

$$\pi = \texttt{Quotient.mk (orbitRel G E)}$$

First things not in mathlib

- Before `instance : ChartedSpace OrbitSpace`, we need some things added to mathlib.
- I created a file: `Mathlib/Topology/Algebra/OrbitSpace.lean`

```
/-!
# Covering and local homeomorphism structure on orbit spaces

Let `G` be a group acting properly discontinuously on a topological space `M`.

This file shows that the quotient map `Quotient.mk : M → orbitRel.Quotient G M`
is a covering map, and hence a local homeomorphism.

We also construct chosen local homeomorphisms and their inverses at each
point, and establish basic properties.
-/
```

The Quotient map is an Homeomorphism

1. $$\left.\begin{array}{l} \pi \text{ is } \texttt{QuotientCoveringMap} \\ \texttt{QuotientCoveringMap} \implies \texttt{CoveringMap} \end{array}\right\} \implies \pi \text{ is } \texttt{CoveringMap}$$

```
lemma isCoveringMap_quotientMk_of_properlyDiscontinuousSMul :
  IsCoveringMap (Quotient.mk _ : M → orbitRel.Quotient G M) :=
 IsQuotientCoveringMap.isCoveringMap (G:=_) (f:=_)
  (isQuotientCoveringMap_quotientMk_of_properlyDiscontinuousSMul)
```

2. $$\left.\begin{array}{l} \pi \text{ is } \texttt{CoveringMap} \\ \texttt{CoveringMap} \overset{\text{(in Mathlib)}}{\implies} \texttt{LocalHomeomorph} \end{array}\right\} \implies \pi \text{ is } \texttt{LocalHomemorph}$$

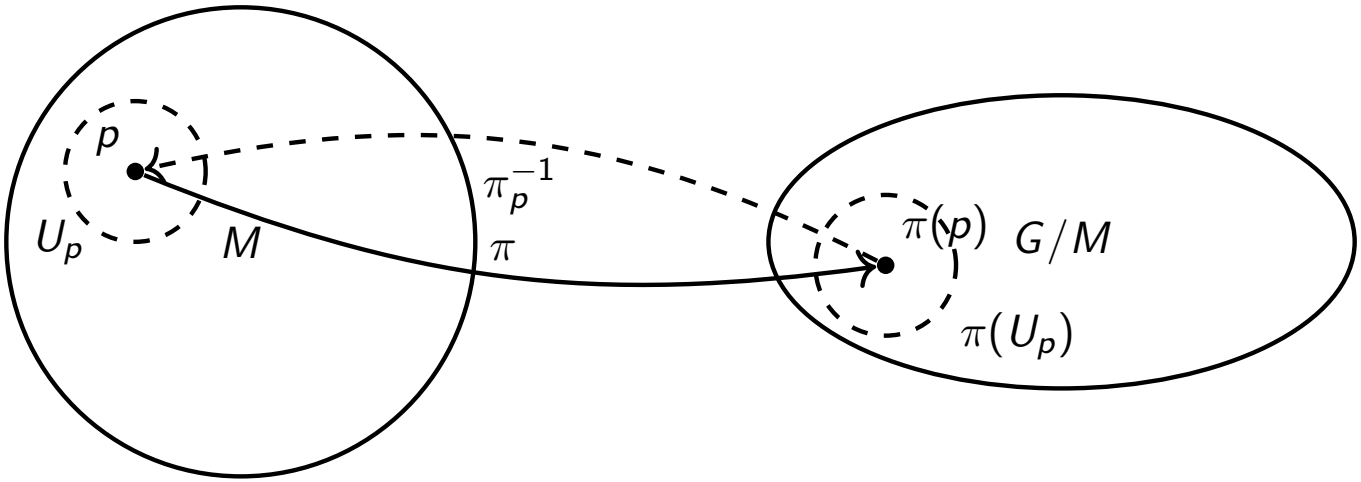
```
lemma isLocalHomeomorph_of_properlyDiscontinuousSMul :
  IsLocalHomeomorph (Quotient.mk _ : M → orbitRel.Quotient G M) :=
 isCoveringMap_quotientMk_of_properlyDiscontinuousSMul
  .isLocalHomeomorph
```

Local homeomorphisms at a point

Definition

$\texttt{IsLocalHomeomorph}\ \ (f : X \to Y)\ \ :=$

$$\forall x \in X, \exists e : \texttt{OpenPartialHomeomorph X Y}, x \in e.\mathit{source} \wedge f = \uparrow e$$




```
noncomputable def localHomeomorphAt (p : M) :
   OpenPartialHomeomorph M (orbitRel.Quotient G M) :=
  Classical.choose (isLocalHomeomorph_of_properlyDiscontinuousSMul p)
```

Basic properties of LocalHomeomorphAt

Recall

$\texttt{IsLocalHomeomorph}\ \ (f : X \to Y)\ \ :=$

$\forall x \in X, \exists e : \texttt{OpenPartialHomeomorph X Y}, x \in e.\mathit{source} \wedge f = \uparrow e$

1. $p \in \pi_p.\mathit{source}$

```
lemma mem_localHomeomorphAt_source {p : M} :
  p ∈ (localHomeomorphAt (G:=G) p).source :=
 (Classical.choose_spec
  (isLocalHomeomorph_of_properlyDiscontinuousSMul p)).1
```

2. $\pi = \uparrow \pi_p$

```
lemma localHomeomorphAt_eq_quotientMk {p : M} :
  localHomeomorphAt (G:=G) p = (Quotient.mk _ : M → orbitRel.Quotient G M)
  :=
 (Classical.choose_spec
  (isLocalHomeomorph_of_properlyDiscontinuousSMul p)).2.symm
```

Extra property of LocalHomeomorphAt

3. The equivalence class $[\![p]\!]$ lies in the target of π_p.

Proof.

- $[\![p]\!] \in (\pi_p).\mathit{target} \iff \exists x \in (\pi_p).\mathit{source}, \pi_p(x) = [\![p]\!] = \pi(p)$
- Take $x = p$, and $p \in (\pi_p).\mathit{source}$ because of Property 1.
- To see $\pi_p(p) = \pi(p)$, see $\pi_p = \pi$ because of Property 2.

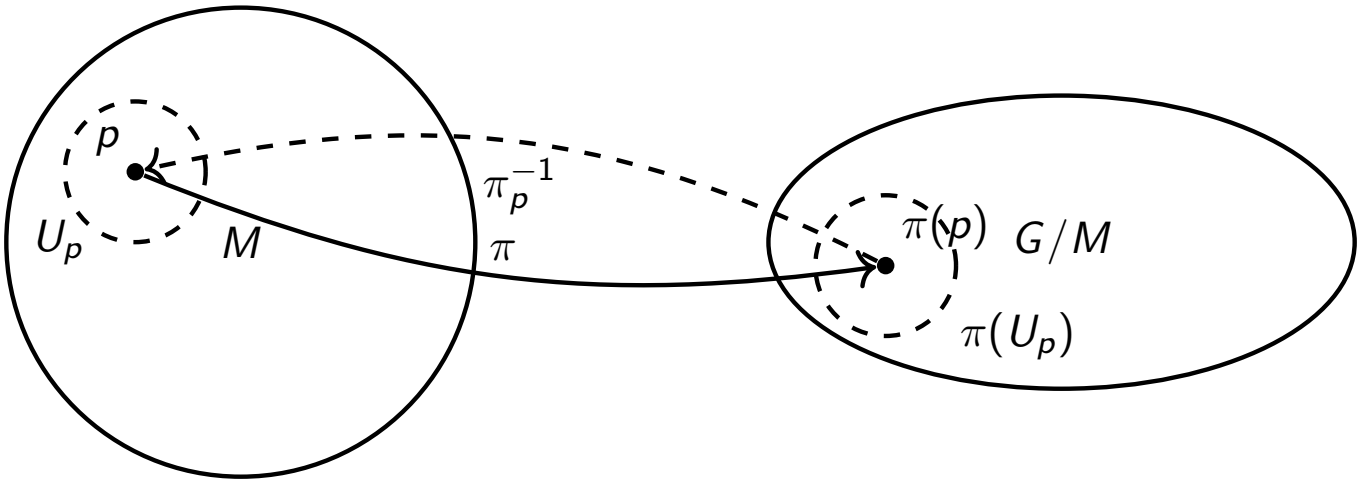
□

```
lemma mem_localHomeomorphAt_target {p : M} :
  ⟦p⟧ ∈ (localHomeomorphAt (G:=G) p).target := by
 rw [← OpenPartialHomeomorph.image_source_eq_target, Set.mem_image]
 refine ⟨p, mem_localHomeomorphAt_source, ?_⟩
 rw [localHomeomorphAt_eq_quotientMk]
```

Local inverses at a point

Since π_p is of type `OpenPartialHomeomorph`, it has an inverse. We will call this inverse $\pi_p^{-1} : M/G \to M$ or `LocalInverseAt p`:

```
def localInverseAt (p : M) : OpenPartialHomeomorph (orbitRel.Quotient G M) M
  := (localHomeomorphAt p).symm
```



Properties of LocalInverseAt

4. The equivalence class $[\![p]\!]$ lies in the source of π_p^{-1}.

Proof.

- $(\pi_p^{-1}).source = (\pi_p).target$.
- By Property 3, $[\![p]\!] \in (\pi_p).target$.

□

```
lemma mem_localInverseAt_source {p : M} :
  ⟦p⟧ ∈ (localInverseAt (G:=G) p).source := by
 simp only [localInverseAt, OpenPartialHomeomorph.symm_source]
 exact mem_localHomeomorphAt_target
```

Properties of LocalInverseAt

5. If $k \in (\pi_p).source$ and $[\![k]\!] \in (\pi_p^{-1}).source$, then $\pi_p^{-1}([\![k]\!]) = k$.

Proof.

- We apply injectivity of π_p on its source, so it suffices $\pi_p(\pi_p^{-1}([\![k]\!])) = \pi_p(k)$.
- We can do this since $k \in (\pi_p).source$ (h) and $\pi_p^{-1}([\![k]\!]) \in (\pi_p).source$ (OPH h').
- Now, $\pi_p(\pi_p^{-1}([\![k]\!])) = [\![k]\!]$ (right inverse of OPH).
- And $\pi_p = \pi$, so $\pi_p(k) = \pi(k) = [\![k]\!]$.

```
lemma localInverseAt_apply_other {p k : M}
   (h : k ∈ (localHomeomorphAt (G := G) p).source)
   (h' : ⟦k⟧ ∈ (localInverseAt (G := G) p).source) :
   (localInverseAt (G := G) p) ⟦k⟧ = k := by
  refine (localHomeomorphAt (G:=G) p).injOn ?_ h ?_
  · simp only [localInverseAt, (localHomeomorphAt p).map_target h']
  · simp only [localInverseAt, (localHomeomorphAt p).right_inv h',
          localHomeomorphAt_eq_quotientMk]
```

Properties of LocalInverseAt

6. The local inverse at p sends $[\![p]\!]$ back to p.

Proof.

- By Property 1, $p \in (\pi_p).source$.
- By Property 4, $[\![p]\!] \in (\pi_p^{-1}).source$.
- Applying Property 5 with $k = p$, we obtain $\pi_p^{-1}([\![p]\!]) = p$.

□

```
lemma localInverseAt_apply_self {p : M}
  : (localInverseAt (G:=G) p) ⟦p⟧ = p :=
 localInverseAt_apply_other
  (mem_localHomeomorphAt_source)
  (mem_localInverseAt_source)
```

Charted space

Definition

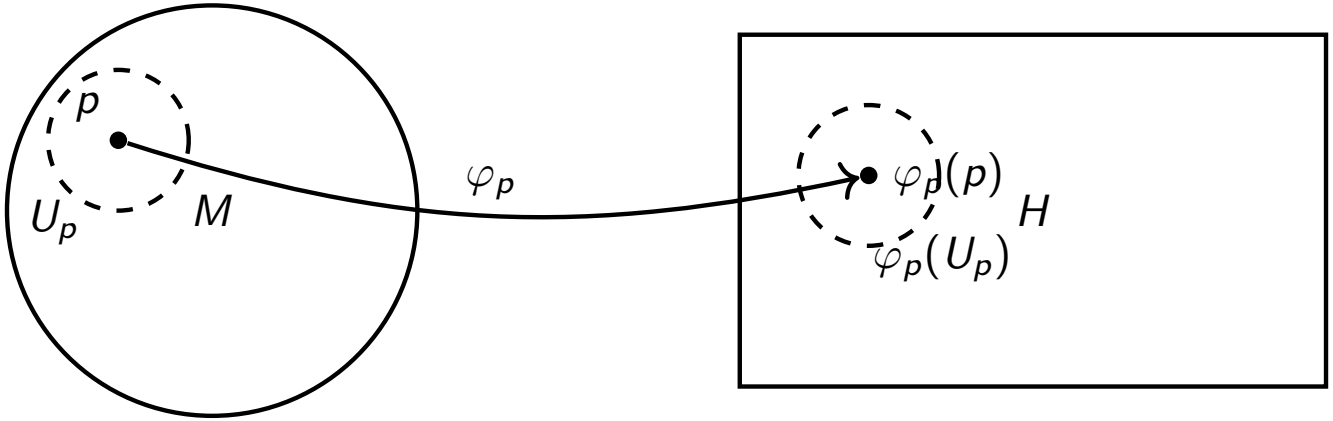
A **charted space** is a topological space M together with a collection of pairs

$$\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$$

such that:

- each $U_i \subseteq M$ is open,
- each $\varphi_i : U_i \to \varphi_i(U_i) \subseteq \mathbb{R}^n$ is a homeomorphism,
- the domains cover M: $\quad M = \bigcup_{i \in I} U_i$.

The pairs (U_i, φ_i) are called **charts**, and $\mathcal{A}$ is called an **atlas**.



ChartedSpace

Instead of requiring that charts cover M, Lean provides a function **choice of chart** at each point.

$$\begin{array}{rcccl} \texttt{chartAt} & : & M & \longrightarrow & \{\text{charts}\} \\ & & p & \longmapsto & \varphi_p \end{array}$$

- $\varphi_p : U_p \to H$ is a local homeomorphism
- $p \in U_p$ (mem_chart_source)
- $\varphi_p \in \mathcal{A}$ (chart_mem_atlas)

This guarantees automatically that the atlas covers M.

```
class ChartedSpace [TopologicalSpace H] [TopologicalSpace M] where
  atlas : Set (OpenPartialHomeomorph M H)
  chartAt : M → OpenPartialHomeomorph M H
  mem_chart_source : ∀ x, x ∈ (chartAt x).source
  chart_mem_atlas : ∀ x, chartAt x ∈ atlas
```

The main result

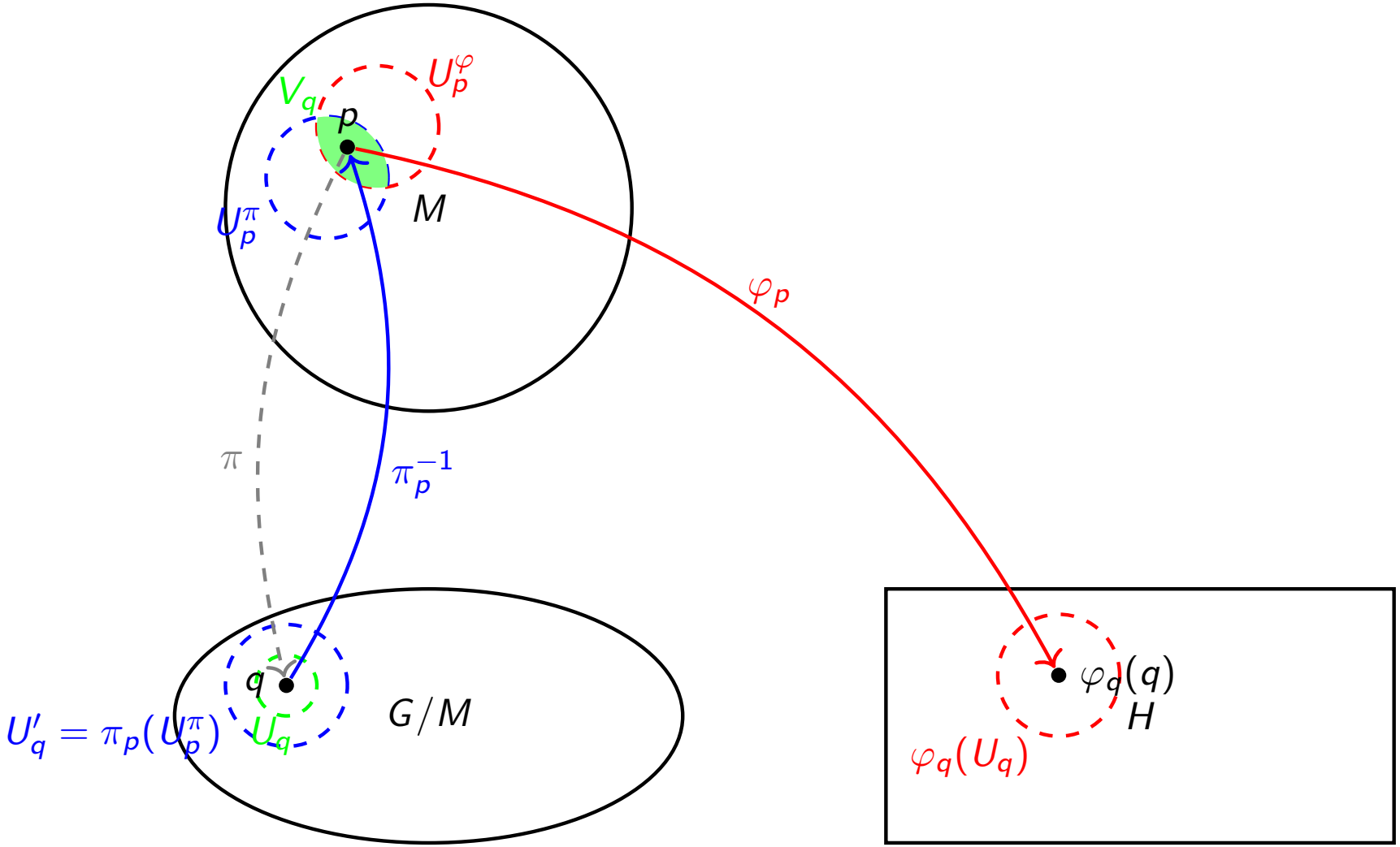
Theorem

Suppose that M is a charted space and G is a group acting properly discontinuously on M. Then the quotient G/M is a charted space.

`Mathlib/Geometry/Manifold/Instances/OrbitSpace.lean`

```
/-!
# Manifold structure on orbit spaces

Let `G` be a group acting properly discontinuously on a manifold `M`.
In this file we endow the orbit space `orbitRel.Quotient G M`
with a manifold structure modeled on the same model space as `M`.

## Main results

Let `M` be a topological space equipped with a `ChartedSpace H M` structure.
Assume that a group `G` acts properly discontinuously on `M`.

Then the orbit space `orbitRel.Quotient G M` inherits a `ChartedSpace H` structure.

## TO-DO

* show that the quotient is an `IsManifold I n` for a suitable
  `ModelWithCorners I`.
* show smoothness of the projection map
```

Construction



Construction

1. Fix $q \in M/G$; we want to construct a chart (U_q, φ_q).
2. Since q is an equivalence class, choose a representative $p \in M$ such that $q = [p]$.
3. Since $\pi_p(p) = q$, the projection has a local inverse near q:
$$\pi_p^{-1} : \pi_p(U_p^\pi) \subset M/G \longrightarrow U_p^\pi \subset M.$$
4. Since M is a charted space, there is a chart at p:
$$\varphi_p : U_p^\varphi \subset M \longrightarrow \varphi_p(U_p^\varphi) \subset H.$$
5. Restricting to a common neighborhood, define
$$\varphi_q = \varphi_p \circ \pi_p^{-1} : U_q \subset M/G \longrightarrow H,$$
where $U_q = \pi_p(U_p^\pi \cap U_p^\varphi)$ is open and contains q.
6. The collection
$$\{(U_q, \varphi_q) \mid q \in M/G\}$$
is an atlas for M/G.

Proof in Lean

```
/-- The orbit space of a properly discontinuous group action on a
manifold inherits a ‘ChartedSpace‘ structure modeled on the same space. -/
instance instChartedSpaceQuotient : ChartedSpace H (orbitRel.Quotient G M)
    where
  atlas := sorry
  chartAt := sorry
  mem_chart_source q := sorry
  chart_mem_atlas := sorry
```

Proof in Lean

q.out = a representative of q

We choose ChartAt $q = \varphi_q = \varphi_p \circ \pi_p^{-1}$

```
/-- The orbit space of a properly discontinuous group action on a
manifold inherits a ‘ChartedSpace‘ structure modeled on the same space. -/
instance instChartedSpaceQuotient : ChartedSpace H (orbitRel.Quotient G M)
   where
 atlas := sorry
 chartAt := fun q => (localInverseAt q.out).trans (chartAt H q.out)
 mem_chart_source q := sorry
 chart_mem_atlas := sorry
```

Proof in Lean

The atlas is then simply the set of all the ChartAts:

$$\mathcal{A} = \{\varphi_q \mid q \in M/G\}$$

So then trivially every `chartAt q` $\in \mathcal{A}$.

```
/-- The orbit space of a properly discontinuous group action on a
manifold inherits a 'ChartedSpace' structure modeled on the same space. -/
instance instChartedSpaceQuotient : ChartedSpace H (orbitRel.Quotient G M)
  where
 atlas := {(localInverseAt q.out).trans (chartAt H q.out) |
                                        q : orbitRel.Quotient G M}
 chartAt := fun q => (localInverseAt q.out).trans (chartAt H q.out)
 mem_chart_source q := sorry
 chart_mem_atlas := by simp
```

Proof in Lean

Lastly we need to prove that for every q, q is in the source of $\varphi_q = \varphi_p \circ \pi_p^{-1}$. This is to say that:

1. q is in the source of π_p^{-1}. Which goes from prop. 4 since $q = [\![p]\!]$.
2. $\pi_p^{-1}(q)$ is in the source of φ_p. But, by prop. 6, we know that $\pi_p^{-1}(q) = \pi_p^{-1}([\![p]\!]) = p$, and p is in the source of φ_p by charted spaceness of M.

```
mem_chart_source q := by
  simp only [OpenPartialHomeomorph.trans_toPartialEquiv,
    PartialEquiv.trans_source, OpenPartialHomeomorph.toFun_eq_coe,
    Set.mem_inter_iff, Set.mem_preimage]
  -- a lot of simplifications; now the goal is
  -- ⊢ q ∈ π_p⁻¹.source ∧ π_p⁻¹ q ∈ φ_p.source
  set p := q.out
  rw [← q.out_eq, localInverseAt_apply_self] -- prop 4
  exact ⟨mem_localInverseAt_source, -- prop 6
        mem_chart_source H p⟩ -- prop of ChartedSpace M
```

Future work

1. Show that the quotient is an `IsManifold I n` for a suitable `ModelWithCorners I`.
2. Prove smoothness of the projection map

$$\pi : M \to M/G.$$

```
instance : IsManifold I n (OrbitSpace M G) := sorry

lemma contMDiff_quotientMk :
   ContMDiff I I n (Quotient.mk _ : M → OrbitSpace M G) := by
  sorry
```

Thank you!

Questions or comments welcome.
Also people who want to help very welcome!

Interesting links:

- My PR #35653
- Quotient Manifolds Project Repo: github.com/grunweg/QuotientManifolds